

1. Administrative & Study Identification

Full Study Title: A Study to Evaluate the Efficacy, Safety, Pharmacokinetics and Dosimetry of [177Lu]Lu-PSMA-617 in Chinese Adult Male Patients With Progressive PSMA-Positive mCRPC **Short Title/Acronym:** [177Lu]Lu-PSMA-617 in Chinese mCRPC **Protocol Number:** CAAA617A12201 **NCT Number:** NCT05670106 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** [177Lu]Lu-PSMA-617 (lutetium vipivotide tetraxetan) and [68Ga]Ga-PSMA-11 **Phase of Development:** Phase II **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy based on the confirmed overall response rate (ORR) of [177Lu]Lu-PSMA-617 when administered in addition to Best Supportive/Best Standard of Care (BSC/BSoc) in Chinese adult male patients with progressive PSMA-positive mCRPC. **Secondary Objectives:** To assess the safety, tolerability, pharmacokinetics (PK), dosimetry, radiographic progression-free survival (rPFS), and overall survival (OS).

2.2 Background & Rationale To address the unmet medical need for effective targeted therapies in patients with mCRPC who have progressed following treatment with taxane regimens and androgen receptor pathway inhibitors (ARPIs). This study provides the first clinical trial evidence of the efficacy, safety, and dosimetry of [177Lu]Lu-PSMA-617 administered with Best Standard of Care (BSoc) specifically in the Chinese population to validate its favorable benefit-risk profile.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Single Group Assignment)

3.2 Planned Enrollment & Duration Target \$N\$: Planned for 60 participants (30 in the main part, 30 in the extension part); 62 participants were ultimately enrolled. **Number of Sites:** Multicenter study across China. **Estimated Study Start:** 01-Jun-2023 **Estimated Primary Completion:** Q1 2024 (Enrollment completed 17-Jan-2024 with subsequent ongoing follow-up)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Chinese male adults ≥ 18 years of age with histological, pathological, and/or cytological confirmation of prostate cancer. 2. Positive [68Ga]Ga-PSMA-11 PET/CT scan, as determined by the sponsor's central reader according to the VISION read rules. 3. Received at least one ARPI (such as enzalutamide and/or abiraterone) and previously treated with at least 1, but no more than 2, prior taxane regimens.

4.2 Exclusion Criteria 1. Previous treatment with Strontium-89, Samarium-153, Rhenium-186, Rhenium-188, Radium-223, or hemi-body irradiation. 2. Previous PSMA-targeted radioligand therapy. 3. Any systemic anti-cancer therapy (e.g., chemotherapy, immunotherapy) or investigational agents within 28 days prior to enrollment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: 7.4 GBq (200 mCi) $\pm$ 10% of [177Lu]Lu-PSMA-617 administered once every 6 weeks. **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Maximum of 6 cycles (up to 36 weeks of active treatment exposure).

5.2 Concomitant & Prohibited Therapy Permitted: Best Supportive/ Best Standard of Care (BSC/BSoc), including ARPIs (e.g., abiraterone, enzalutamide), as determined by the local investigator's clinical best practice. **Prohibited:** Other concurrent cytotoxic chemotherapy, immunotherapy, external radioligand therapy, or investigational therapy.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Eligibility, [68Ga]Ga-PSMA-11 PET/CT scan (within 21 days of enrollment).
Baseline	Day 0	Enrollment, First dose of [177Lu]Lu-PSMA-617.
Interim	Every 6 Weeks	IV Administration of [177Lu]Lu-PSMA-617, Safety Labs, PK/dosimetry evaluations.
Imaging Follow-up	Weeks 8, 16, 24; then Q12W	Radiographic imaging every 8 weeks for the first 24 weeks, then every 12 weeks thereafter.
End of Study	EOT Visit	Final vital signs, safety assessments, imaging (if not done within 28 days), and survival tracking.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Confirmed Overall Response Rate (ORR). **Measurement Method:** Independent centralized review of

radiographic images utilizing the PCWG3-modified RECIST v1.1 criteria.

7.2 Secondary & Exploratory Endpoints - Radiographic progression-free survival (rPFS) and Overall Survival (OS). - Blood pharmacokinetics (PK) and organ dosimetry of [177Lu]Lu-PSMA-617 and [68Ga]Ga-PSMA-11. - Safety and tolerability evaluations, identifying specific frequencies of treatment-emergent myelosuppression, hepatotoxicity, and renal toxicity.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection combined with rigorous monitoring of laboratory data (hematology and chemistry) to track known radioligand AE profiles (e.g., myelosuppression). **Serious Adverse Events (SAEs):** Routine monitoring and reporting framework, typically mandating 24-hour notification. **Data Monitoring Committee (DMC):** Managed centrally under Novartis Pharmaceuticals clinical operations.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: The Main part cohort ($n=30$ participants with measurable lesions) was leveraged to analyze the primary confirmed ORR endpoint. All participants (Main + Extension parts) were utilized for secondary safety and efficacy endpoints. **Sample Size Justification:** A sample size of ~60 participants, combined with an intensive subset of 10-12 participants for PK/dosimetry, optimally powered safety and efficacy determinations in the target population. **Handling of Missing Data:** Standard imputation logic as strictly dictated by the PCWG3-modified RECIST v1.1 criteria.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-cadence site monitoring alongside rigorous centralized radiographic reads by independent reviewers. **Audit Trail:** Secure electronic Case Report Forms (eCRF) utilizing automated data verification algorithms and comprehensive documentation logs.

11. Study Identifiers & Governance

Primary ID: CAAA617A12201 **Registry Number:** NCT05670106 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** A Study to Evaluate the Efficacy, Safety, Pharmacokinetics and Dosimetry

of [177Lu]Lu-PSMA-617 in Chinese Adult Male Patients With Progressive PSMA-Positive mCRPC **Official Regulatory Title:** A Study to Evaluate the Efficacy, Safety, Pharmacokinetics and Dosimetry of [177Lu]Lu-PSMA-617 in Chinese Adult Male Patients With Progressive PSMA-Positive mCRPC

12. Clinical Framework (mCRPC Specific)

Primary Condition: Metastatic Castration-Resistant Prostate Cancer (mCRPC) **Diagnosis Threshold:** PSMA-positive mapping confirmed via [68Ga]Ga-PSMA-11 PET/CT **Medical Methodology:** PSMA-Targeted Radioligand Therapy alongside Best Supportive Care (BSC) **Optimization Target:** Radiographic tumor response reduction and durable progression-free survival (rPFS).

13. Study Procedures & Methodology

Management Pathway: Specialized nuclear medicine administration of radioactive therapeutic compounds **Education Protocol (HCP):** Stringent institutional guidelines around radioactive material dosing, disposal, and dosimetry tracking. **Education Protocol (Patient):** Radiation safety precautions following therapeutic infusion and AE recognition (myelosuppression signs) **Quality Audit Mechanism:** Independent centralized radiographic image analysis confirming PCWG3-modified RECIST v1.1 parameters

14. Observation & Follow-Up Schedule

Total Duration: Ongoing until radiographic disease progression, death, or participant withdrawal **Onsite Visit Cadence:** Every 6 weeks for radioligand dosing (Cycles 1-6). **Remote/Imaging Monitoring Frequency:** Radiographic CT/MRI scans administered every 8 weeks for the first 24 weeks, followed by every 12 weeks **Checklist Review Interval:** Routine safety lab evaluation before the commencement of every treatment cycle.

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator		Dingwei Ye		Signed		03-Apr-2026										
	Clinical Operations Lead		Novartis Clinical Operations Lead		Signed		03-Apr-2026			Lead Statistician		Novartis Lead					
	Statistician		Signed		03-Apr-2026												